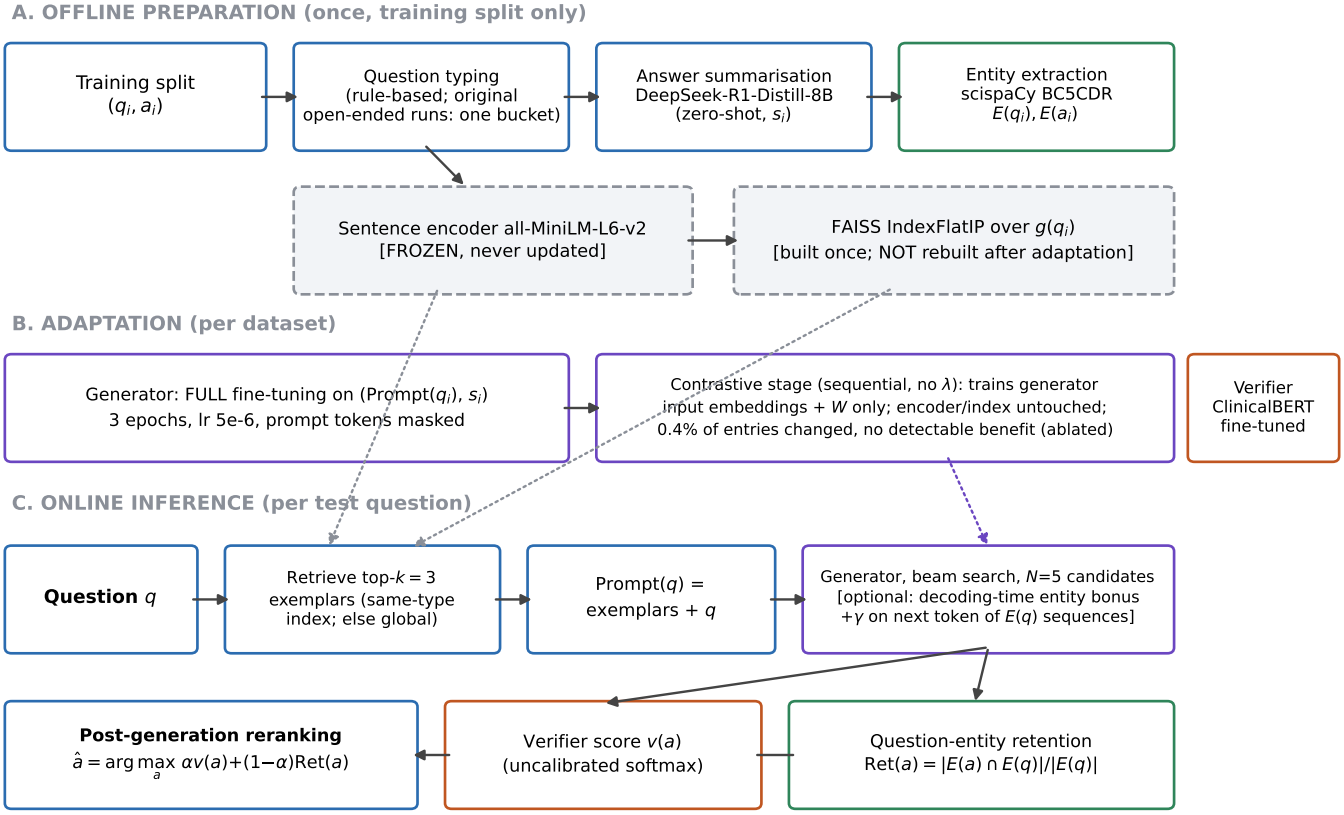
A. OFFLINE PREPARATION (once, training split only)
Training split (q_i, a_i)
Question typing (rule-based; original open-ended runs: one bucket)
Answer summarisation DeepSeek-R1-Distill-8B (zero-shot, s_i)
Entity extraction scispaCy BC5CDR $E(q_i)$, $E(a_i)$
Sentence encoder all-MiniLM-L6-v2 [FROZEN, never updated]
FAISS IndexFlatIP over $g(q_i)$ [built once; NOT rebuilt after adaptation]
B. ADAPTATION (per dataset)
Generator: FULL fine-tuning on (Prompt(q_i), s_i) 3 epochs, lr 5e-6, prompt tokens masked
Contrastive stage (sequential, no λ): trains generator input embeddings + W only; encoder/index untouched; 0.4% of entries changed, no detectable benefit (ablated)
Verifier ClinicalBERT fine-tuned
C. ONLINE INFERENCE (per test question)
Question q
Retrieve top-$k = 3$ exemplars (same-type index; else global)
Prompt(q) = exemplars + q
Generator, beam search, N=5 candidates [optional: decoding-time entity bonus $+\gamma$ on next token of $E(q)$ sequences]
Post-generation reranking
$\hat{a} = \arg\max_a \alpha v(a)+(1-\alpha)\mathrm{Ret}(a)$
Verifier score $v(a)$ (uncalibrated softmax)
Question-entity retention
$\mathrm{Ret}(a) = |E(a) \cap E(q)|/|E(q)|$